

Question bank for Unit 5

Principles of Quantum Computing.

1. What is the primary advantage of quantum computing over classical computing?
2. Which principle allows a qubit to exist in multiple states simultaneously?
3. What is the phenomenon where qubits become correlated, regardless of distance?
4. What is the effect of the Hadamard gate on the basis state $|1\rangle$ in Dirac notation?
5. Which gate swaps the two qubits in a quantum circuit?
6. Calculate the effect of the Pauli X matrix on an arbitrary qubit given by $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.
7. Calculate the output of CNOT gate on the state $|00\rangle$ using the matrix representation.
8. If a qubit is in the state $|\psi\rangle = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$. Compute the probability of measuring $|0\rangle$.
9. What is the orthonormality of the basis states?
10. With necessary diagram elucidate Charge Qubit.
11. Describe the Pauli X, Y and Z gates and their operations on $|\psi\rangle$.
12. Describe Briefly on Quantum Harmonic Oscillator analogous to LC oscillator. Explain the need of anharmonicity.
13. State the matrix representation of a) Hadamard Gate, b) CNOT gate. Explain their action in Dirac Notation. Write down their Truth Tables.
14. Explain the working of T gate with matrix and truth table.
15. Explain the representation of Qubit by Bloch sphere.

Numericals:

1. Show that Hadamard gate is Unitary.

Soln:

$$H^\dagger H = I$$

2. Check the normality of the Qubit $|\psi\rangle = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$.

$$\|\psi\| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{25}{25}} = \sqrt{1} = 1$$

The normalized is equal to 1, Hence it is normalized qubit state.

3. Verify if the matrix $M = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$ is Hermitian

$$M^\dagger = (\overline{M})^T = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$$

$$M^\dagger = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} = M$$

Yes, M is **Hermitian** because $M^\dagger = M$

4. Calculate the action of Pauli X gate followed by Hadamard gate on the state $\alpha|0\rangle + \beta|1\rangle$

$$X|\psi\rangle = \beta|0\rangle + \alpha|1\rangle.$$

$$HX(\alpha|0\rangle + \beta|1\rangle) = \frac{1}{\sqrt{2}} [(\alpha + \beta)|0\rangle + (\beta - \alpha)|1\rangle]$$